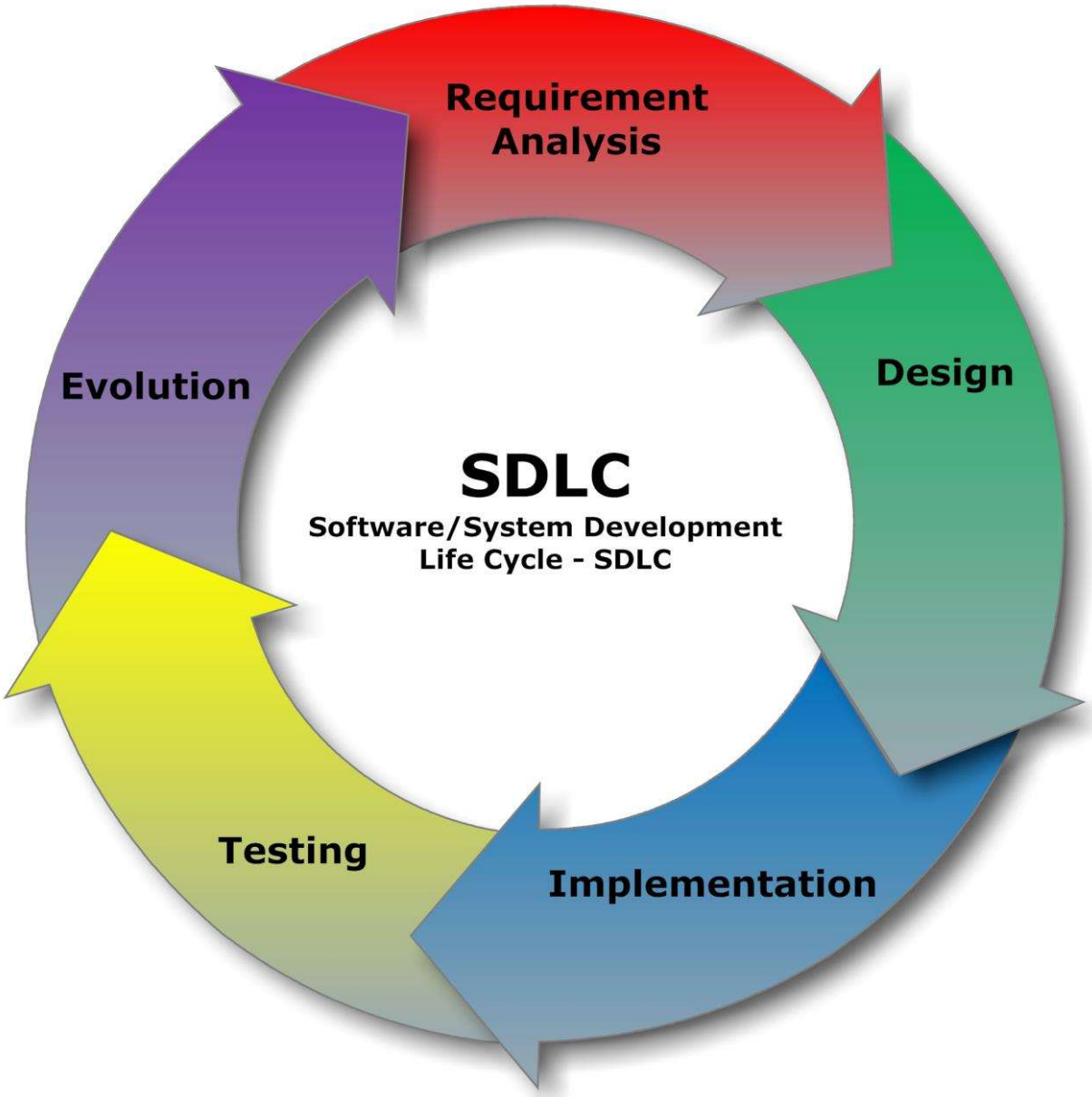
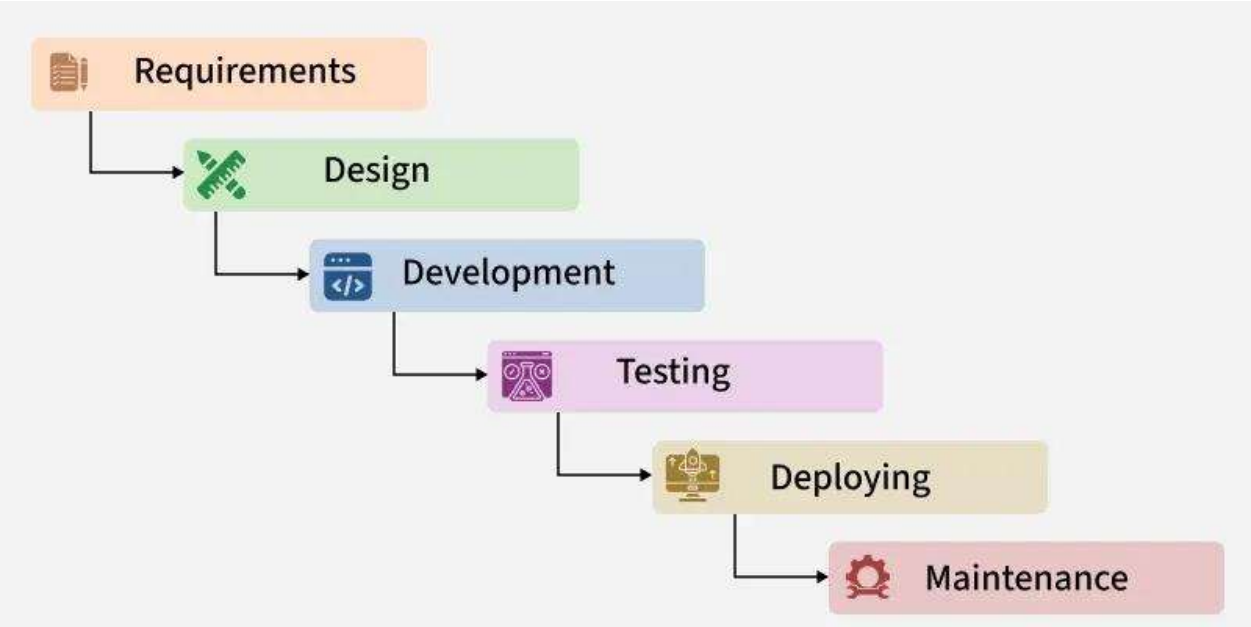
SDLC

(Software Development Life Cycle)



Waterfall Model in SDLC-

The **Waterfall Model** is one of the earliest and most traditional **Software Development Life Cycle (SDLC)** approaches, introduced by *Winston W. Royce* in 1970. It follows a **linear and sequential** process where each phase must be completed before moving to the next, making it ideal for projects with **well-defined and stable requirements**.



Drawback of SDLC Waterfall Models-

- **Rigid and Sequential Process** Each phase (requirements → design → implementation → testing → deployment → maintenance) must be completed before moving to the next. This rigidity makes it unsuitable for projects where requirements evolve.

- **Late Discovery of Issues** Testing only happens after development is complete. Bugs or design flaws are often discovered very late, making fixes expensive and time-consuming.

- **Poor Adaptability to Change** Once requirements are documented, changes are difficult to incorporate. In real-world projects, requirements often shift due to user feedback or market changes.

- **High Risk of Requirement Misunderstanding** If requirements are misunderstood at the start, the entire project may fail because there's no mechanism for continuous feedback during development.

- **Limited User Involvement** Users are typically involved only at the beginning (requirements of gathering) and end (delivery). This can lead to dissatisfaction if the final product doesn't meet expectations.

- **Unsuitable for Complex or Long-Term Projects** For large projects, the Waterfall model struggles because it cannot handle uncertainty or evolving technologies effectively.